

Chapter 20

Nonlinear Wave Generation and C & G Harmonic Scatter-Field Dynamics

Mathematical formulation for transverse waves, nonlinear harmonic generation, string modes, Lorentz covariance and curved spacetime

20.1 Scope and scientific status

The established mathematics in this chapter comes from classical nonlinear-wave theory, relativistic field theory, general relativity and classical string dynamics. The C & G terms are introduced as a proposed phenomenological extension. They do not constitute an established physical law unless a complete action, dimensional analysis, boundary conditions and testable predictions are supplied.

20.2 Linear transverse-wave foundation

Let $y(x,t)$ denote a displacement perpendicular to the direction of propagation x . For a uniform string of tension T and linear mass density μ , force balance gives

$$\mu \partial^2 y / \partial t^2 = T \partial^2 y / \partial x^2 \quad (20.1)$$

$$\partial^2 y / \partial t^2 - c^2 \partial^2 y / \partial x^2 = 0, \quad c = \sqrt{T/\mu} \quad (20.2)$$

A monochromatic solution is

$$y(x,t) = A \cos(kx - \omega t + \varphi) \quad (20.3)$$

$$\omega = ck, \quad k = 2\pi/\lambda, \quad \omega = 2\pi f, \quad c = f\lambda \quad (20.4)$$

The transverse particle velocity and local slope are

$$v_y = \partial y / \partial t = A\omega \sin(kx - \omega t + \varphi) \quad (20.5)$$

$$\partial y / \partial x = -Ak \sin(kx - \omega t + \varphi) \quad (20.6)$$

The time-averaged energy density of the ideal string is

$$\langle u \rangle = \frac{1}{2} \mu \omega^2 A^2 \quad (20.7)$$

20.3 Origin of nonlinearity

Linear theory assumes small slope $|\partial y / \partial x| \ll 1$. At finite amplitude, geometric, elastic or material effects make the restoring force amplitude dependent. A general one-dimensional model is

$$\partial^2 y / \partial t^2 - c^2 \partial^2 y / \partial x^2 + 2\gamma \partial y / \partial t + \alpha y + \beta y^2 + \chi y^3 = S(x,t) \quad (20.8)$$

Here γ is damping, α is the linear restoring coefficient, β and χ are quadratic and cubic nonlinearities, and S is an external source. In a spatially dispersive medium one may also include higher derivatives:

$$\partial^2 y / \partial t^2 - c^2 \partial^2 y / \partial x^2 + d_4 \partial^4 y / \partial x^4 + N[y] = S \quad (20.9)$$

20.4 Harmonic generation

Suppose the system is driven at angular frequency ω with a leading-order response

$$y_1 = A \cos \theta, \quad \theta = kx - \omega t \quad (20.10)$$

The quadratic term generates a constant component and a second harmonic because

$$y_1^2 = A^2 \cos^2 \theta = \frac{1}{2} A^2 [1 + \cos(2\theta)] \quad (20.11)$$

The cubic term generates corrections at the fundamental and a third harmonic:

$$y_1^3 = A^3 \cos^3 \theta = \frac{1}{4} A^3 [3 \cos \theta + \cos(3\theta)] \quad (20.12)$$

Therefore a perturbative field can be written

$$y \approx A_1 \cos \theta + A_2 \cos(2\theta + \varphi_2) + A_3 \cos(3\theta + \varphi_3) + y_{\text{DC}} \quad (20.13)$$

For weak nonlinearity, the generated amplitudes scale approximately as

$$A_2 \propto \beta A_1^2, \quad A_3 \propto \chi A_1^3 \quad (20.14)$$

20.5 Two-mode C & G resonance hypothesis

Represent the proposed C and G components as two transverse modes:

$$y_{\text{C}} = A_{\text{C}} \cos(k_{\text{C}}x - \omega_{\text{C}}t + \varphi_{\text{C}}) \quad (20.15)$$

$$y_{\text{G}} = A_{\text{G}} \cos(k_{\text{G}}x - \omega_{\text{G}}t + \varphi_{\text{G}}) \quad (20.16)$$

$$y_{\text{CG}} = y_{\text{C}} + y_{\text{G}} \quad (20.17)$$

For a proposed perfect-fifth frequency relation,

$$\omega_{\text{G}} / \omega_{\text{C}} = f_{\text{G}} / f_{\text{C}} = 3/2 \quad (20.18)$$

A phase-locking measure is

$$\Gamma_{\text{CG}} = |\langle \exp[i(\varphi_{\text{G}} - 3\varphi_{\text{C}}/2)] \rangle|, \quad 0 \leq \Gamma_{\text{CG}} \leq 1 \quad (20.19)$$

Because a 3:2 ratio is not a one-to-one phase identity, a more exact periodic-locking condition uses integers $p=2$ and $q=3$:

$$\Delta\Phi_{2:3} = 2\varphi_{\text{G}} - 3\varphi_{\text{C}}, \quad \Gamma_{2:3} = |\langle \exp(i\Delta\Phi_{2:3}) \rangle| \quad (20.20)$$

Perfect modal locking corresponds ideally to $\Gamma_{2:3} = 1$. In real systems damping, detuning and noise generally give $\Gamma_{2:3} < 1$.

20.6 Nonlinear C-G coupling model

A minimal coupled-mode model is

$$\ddot{y}_C + 2\gamma_C \dot{y}_C + \omega_C^2 y_C + a_C y_C^3 + \kappa y_C y_G^2 = F_C \cos(\Omega_C t) \quad (20.21)$$

$$\ddot{y}_G + 2\gamma_G \dot{y}_G + \omega_G^2 y_G + a_G y_G^3 + \kappa y_G y_C^2 = F_G \cos(\Omega_G t) \quad (20.22)$$

The interaction energy associated with the cross-coupling may be written

$$V_{\text{int}} = \frac{1}{2} \kappa y_C^2 y_G^2 \quad (20.23)$$

The equations follow from the effective Lagrangian

$$L = \frac{1}{2} \dot{y}_C^2 + \frac{1}{2} \dot{y}_G^2 - \frac{1}{2} \omega_C^2 y_C^2 - \frac{1}{2} \omega_G^2 y_G^2 - \frac{1}{4} a_C y_C^4 - \frac{1}{4} a_G y_G^4 - \frac{1}{2} \kappa y_C^2 y_G^2 \quad (20.24)$$

The nonlinear frequency shift of a weak Duffing mode is approximately

$$\Delta\omega_C \approx (3a_{CA_C^2})/(8\omega_C), \quad \Delta\omega_G \approx (3a_{GA_G^2})/(8\omega_G) \quad (20.25)$$

20.7 Scatter-wave field

Let an incident transverse field interact with N scattering centres. The total complex field is

$$\Psi_{\text{tot}}(r,t) = \Psi_{\text{inc}}(r,t) + \sum_{j=1}^N \Psi_j(r,t) \quad (20.26)$$

$$\Psi_j(r,t) = [f_j(\theta, \varphi)/r_j] A_j \exp[i(k r_j - \omega t + \delta_j)] \quad (20.27)$$

The measured intensity contains interference terms:

$$I = |\Psi_{\text{tot}}|^2 = |\Psi_{\text{inc}}|^2 + \sum_j |\Psi_j|^2 + 2\text{Re}[\Psi_{\text{inc}}^* \sum_j \Psi_j] + 2\text{Re}[\sum_{i<j} \Psi_i^* \Psi_j] \quad (20.28)$$

A nonlinear scatterer can generate new frequencies. For an input with frequencies ω_1 and ω_2 , quadratic mixing produces

$$\omega_{\text{out}} \in \{0, 2\omega_1, 2\omega_2, \omega_1 + \omega_2, |\omega_1 - \omega_2|\} \quad (20.29)$$

Cubic mixing additionally produces components such as

$$\omega_{\text{out}} \in \{3\omega_1, 3\omega_2, 2\omega_1 \pm \omega_2, 2\omega_2 \pm \omega_1\} \quad (20.30)$$

20.8 Resonance distillation

“Distillation” may be defined mathematically as projection of a measured field onto selected coherent modes. With inner product

$$\langle u, v \rangle = \int_V u^*(r,t) v(r,t) dV \quad (20.31)$$

the modal coefficient is

$$a_n(t) = \langle u_n, \Psi_{\text{tot}} \rangle / \langle u_n, u_n \rangle \quad (20.32)$$

The distilled C & G field is

$$\Psi_{\text{CG}}^{\text{(dist)}} = a_C u_C + a_G u_G \quad (20.33)$$

A coherence-weighted form is

$$\Psi_{CG}^{(coh)} = \Gamma_{2;3} [a_{Cu_C} + a_{Gu_G}] \quad (20.34)$$

20.9 Relativistic consistency

A proposed fundamental field must be written covariantly. For a scalar field H in flat spacetime,

$$\square H + dV/dH = J, \quad \square = \eta^{\{\mu\nu\}} \partial_\mu \partial_\nu = -c^2 \partial^2 / \partial t^2 + \nabla^2 \quad (20.35)$$

The Lorentz transformation is

$$x' = \gamma_L(x - vt), \quad t' = \gamma_L(t - vx/c^2), \quad \gamma_L = 1/\sqrt{1-v^2/c^2} \quad (20.36)$$

Lorentz covariance requires the action to be a spacetime scalar. One possible scalar-field action is

$$S_H = \int d^4x [-\frac{1}{2} \partial_\mu H \partial^\mu H - V(H) + JH] \quad (20.37)$$

The symbol γ_L denotes the Lorentz factor and is distinct from the damping coefficient γ used earlier.

20.10 Curved-spacetime formulation and General Relativity

On curved spacetime, replace the flat d'Alembertian by

$$\square_g H = (1/\sqrt{-g}) \partial_\mu (\sqrt{-g} g^{\{\mu\nu\}} \partial_\nu H) \quad (20.38)$$

The nonlinear curved-spacetime field equation is

$$\square_g H - dV/dH = -J \quad (20.39)$$

For a quartic potential,

$$V(H) = \frac{1}{2} m_H^2 H^2 + (\lambda_3/3!) H^3 + (\lambda_4/4!) H^4 \quad (20.40)$$

$$\square_g H - m_H^2 H - (\lambda_3/2) H^2 - (\lambda_4/6) H^3 = -J \quad (20.41)$$

Einstein's field equation is

$$G_{\{\mu\nu\}} + \Lambda g_{\{\mu\nu\}} = (8\pi G_N/c^4) T_{\{\mu\nu\}} \quad (20.42)$$

If H is a physical field, its stress-energy contribution follows from variation of its action:

$$T_{\{\mu\nu\}}^{(H)} = \partial_\mu H \partial_\nu H - g_{\{\mu\nu\}} [\frac{1}{2} \partial_\alpha H \partial^\alpha H + V(H)] \quad (20.43)$$

$$G_{\{\mu\nu\}} + \Lambda g_{\{\mu\nu\}} = (8\pi G_N/c^4) (T_{\{\mu\nu\}}^{(matter)} + T_{\{\mu\nu\}}^{(H)}) \quad (20.44)$$

Equations (20.43)–(20.44) are a standard scalar-field construction. Identifying H with a C & G harmonic field is a hypothesis and requires empirical justification.

20.11 Nonlinear string modes

For a relativistic string with world-sheet coordinates (τ, σ) , the Polyakov action is

$$S_P = -(T_s/2) \int d\tau d\sigma \sqrt{-h} h^{ab} \partial_a X^\mu \partial_b X_\mu \quad (20.45)$$

In conformal gauge the linear string equation is

$$(\partial^2/\partial\tau^2 - \partial^2/\partial\sigma^2)X^\mu = 0 \quad (20.46)$$

A phenomenological nonlinear transverse string model may be written

$$(\partial_\tau^2 - \partial_\sigma^2)Y + \alpha_s Y^2 + \beta_s Y^3 + \kappa_s Y_{CY} G^2 = 0 \quad (20.47)$$

This nonlinear equation is not the fundamental critical superstring equation; it is an effective classical model for exploring mode coupling on a string-like medium.

20.12 Dimensionless formulation

Let $x=L\xi$, $t=(L/c)\tau$ and $y=Y_0u$. Then equation (20.8) becomes

$$u_{\{\tau\tau\}} - u_{\{\xi\xi\}} + 2\zeta u_\tau + au + bu^2 + qu^3 = s(\xi, \tau) \quad (20.48)$$

with dimensionless coefficients

$$\zeta=\gamma L/c, \quad a=\alpha L^2/c^2, \quad b=\beta Y_0 L^2/c^2, \quad q=\chi Y_0^2 L^2/c^2 \quad (20.49)$$

This form makes the controlling ratios explicit and is preferable for numerical simulation and parameter comparison.

20.13 Stability and resonance criteria

For a spatially uniform equilibrium $y=y_0$, stability requires the effective curvature of the potential to be positive:

$$V_{\text{eff}}''(y_0) = \alpha + 2\beta y_0 + 3\chi y_0^2 > 0 \quad (20.50)$$

For a driven damped mode, the approximate steady-state amplitude A obeys

$$A^2[(\omega_0^2 - \Omega^2 + \frac{3}{4}\chi A^2)^2 + (2\gamma\Omega)^2] = F^2 \quad (20.51)$$

Nonlinear resonance can therefore bend, broaden, split or become multivalued. “Perfect resonance” is an ideal limit; real systems are bounded by damping, bandwidth, detuning and instability.

20.14 Testable quantities

- Harmonic conversion efficiencies $\eta_2=P(2\omega)/P(\omega)$ and $\eta_3=P(3\omega)/P(\omega)$.
- Phase-locking statistic $\Gamma_{2:3}$ and its dependence on drive strength.
- Nonlinear frequency shifts $\Delta\omega_C$ and $\Delta\omega_G$.
- Scattering cross-sections and angular distributions at generated frequencies.
- Damping rates, quality factors $Q=\omega_0/(2\gamma)$, and coherence times.
- Any deviation predicted by the proposed C & G coupling relative to the baseline model $\kappa=0$.

20.15 Consolidated governing equation

A compact phenomenological equation combining transverse propagation, dispersion, damping, nonlinear generation, external forcing and a proposed C & G coupling is

$$y_{\{tt\}} - c^2 y_{\{xx\}} + d_4 y_{\{xxxx\}} + 2\gamma y_t + \alpha y + \beta y^2 + \chi y^3 + \kappa H_{CG} y = S_{scatter}(x,t) \quad (20.52)$$

For relativistic or gravitational applications, the corresponding covariant form is

$$\square_g H - m_H^2 H - (\lambda_3/2) H^2 - (\lambda_4/6) H^3 - \kappa_{CG} \mathcal{C} \mathcal{G} H = -J_{scatter} \quad (20.53)$$

where $\mathcal{C}$ and $\mathcal{G}$ must be defined as covariant scalar quantities, or replaced by an explicitly tensorial coupling. Without that definition, equation (20.53) remains a schematic research ansatz rather than a complete theory.

20.16 Conclusion

Nonlinear wave generation provides the appropriate mathematical language for amplitude-dependent resonance, frequency conversion, mode coupling and scatter-field redistribution. The C & G proposal can be represented as a coupled-mode or covariant-field hypothesis, but scientific assessment requires dimensional consistency, a specified action, well-posed initial and boundary conditions, and predictions distinguishable from established nonlinear physics.